Appendix

1 Appendix

1.1 Optional Runtime Roadmap Personalization

RAS-FL uses the global runtime roadmap by default. However, a deployment client with a small local adaptation set may personalize the roadmap to better reflect its data distribution. This process modifies only the importance ordering and selected structures; the shared model parameters $\mathbf{w}^*$ remain fixed. Let $\mathcal{D}_u^a$ denote the adaptation set of deployment client u. Using the fixed global model, the client estimates local structure importance as

$$\mathbf{a}_u^l = \Phi(\mathbf{w}^*; \mathcal{D}_u^a)\,, \tag{1}$$

where $\Phi(\cdot)$ is the importance-estimation procedure described in **Stage II**. The local estimate is combined with the global importance vector:

$$\mathbf{a}_u^{\mathrm{pers}} = \lambda_u \mathbf{a}^{g,*} + (1 - \lambda_u)\mathbf{a}_u^l, \tag{2}$$

where

$$\lambda_u = \frac{1}{1 + \gamma|\mathcal{D}_u^a|}. \tag{3}$$

Thus, clients with limited adaptation data rely primarily on the global importance ordering, while larger adaptation sets permit stronger local personalization. The personalized importance vector is used to construct

$$\mathcal{R}_u = \{\mathbf{m}_u^p : p \in \{0\} \cup \mathcal{P}\}\,, \tag{4}$$

using the same removal budgets, structured masking procedure, and layer-wise boundary constraints as the global roadmap. Personalization therefore changes only which structures are retained at each shrinkage level and does not update $\mathbf{w}^*$. When no adaptation data are available, the client uses

$$\mathcal{R}_u = \mathcal{R}^g, \tag{5}$$

which remains the default deployment mode.

Default and personalised deployment. RAS-FL primarily targets global-roadmap deployment, which requires no client-side adaptation and generally provides the strongest and most consistent performance. Personalisation is presented as an optional extension for clients with local adaptation data. Although it may not outperform the global roadmap, it preserves comparable accuracy at most operating points, typically remaining within one percentage point, while modifying only the retained structures and keeping $\mathbf{w}^*$ fixed. Thus, the global roadmap remains the default, whereas personalisation demonstrates that client-specific adaptation is available without retraining or fine-tuning the shared model.

Algorithm 1 Runtime-Aware Shrinkable FL (RAS-FL)

Require: Training clients $\mathcal{K}$, rounds T, local steps E, shrinkage levels $\mathcal{P}$, β, and $\rho_{\min}$
Ensure: $\mathbf{w}^*$, $\mathbf{a}^{g,*}$, and $\mathcal{R}^g$
Initialize $\mathbf{w}^0$ and $\mathbf{a}^{g,0} = \epsilon\mathbf{1}$ using Eq.(**6**)
for $t = 0, 1, \ldots, T-1$ **do**
  Select $\mathcal{S}_t \subseteq \mathcal{K}$
  if shrinkage-aware training is active **then**
    Construct $\{\mathbf{m}^{p,t} : p \in \mathcal{P}_t\}$ from $\mathbf{a}^{g,t}$ using the boundary-aware roadmap procedure
  end if
  Broadcast $\mathbf{w}^t$
  for each client $k \in \mathcal{S}_t$ **in parallel do**
    Set $\mathbf{w}_k^{t,0} \leftarrow \mathbf{w}^t$
    Compute local importance using Eq.(**7**)
    for $e = 0, 1, \ldots, E-1$ **do**
      if shrinkage-aware training is active and $z \sim \mathrm{Bernoulli}(\pi)$ yields $z = 1$ **then**
        Sample $p \in \mathcal{P}_t$
        Optimize Eq.(**16**)
        Apply Eq.(**17**)
      else
        Optimize the full-model objective (first case of Eq.(**16**))
      end if
    end for
    Upload $(\mathbf{w}_k^{t,E}, \mathbf{a}_k^t)$
  end for
  Aggregate model parameters and importance using Eq.(**10**)
  Update the global importance vector using Eq.(**11**)
  Compute S^t using Eq.(**19**)
end for
Select t^* using Eq.(**20**)
Obtain $\mathbf{w}^*$ and $\mathbf{a}^{g,*}$ using Eq.(**21**)
Construct $\mathcal{R}^g$ using Eq.(**28**)
return $\mathbf{w}^*$, $\mathbf{a}^{g,*}$, and $\mathcal{R}^g$

1.2 Baseline Implementations

We adapted all baselines to the same federated training and deployment setting used by RAS-FL. The methods use identical client partitions, communication rounds, local epochs, optimizer settings, and evaluation clients.**FedAvg** and **FedProx** are implemented as full-model FL baselines. Each client trains the complete network, the server aggregates the resulting model updates, and deployment always uses the full model. FedProx additionally includes its proximal regularization term during local optimization.

FedDrop-style is implemented as a random structured-submodel baseline. At each supported shrinkage level, filters, channels, or neurons are randomly deactivated while enforcing the same layer-wise minimum-capacity constraints, protected layers, and global removal budgets used by RAS-FL. This baseline therefore isolates the effect of importance-guided structure selection from the effect of structured shrinkage itself.

Algorithm 2 Runtime Deployment on a New Client

Require: $\mathbf{w}^*$, $\mathbf{a}^{g,*}$, $\mathcal{R}^g$, client u, adaptation data $\mathcal{D}_u^a$, and resource state $\mathbf{r}_u(t)$
Ensure: Prediction $\hat{y}$

if $\mathcal{D}_u^a$ is available **then**
    Compute $\mathbf{a}_u^l$, λ_u, and $\mathbf{a}_u$ using Eqs. (1)–(3)
    Construct $\mathcal{R}_u$ using Eq. (4)
else
    Set $\mathcal{R}_u \leftarrow \mathcal{R}^g$
end if
Select $p_u(t)$ using Eq.**(28)**
Retrieve $\mathbf{m}_u^{p_u(t)}$ from $\mathcal{R}_u$
Release the currently active GPU model
Construct $\mathbf{w}_c^{p_u(t)} \leftarrow \mathcal{C}(\mathbf{w}^*, \mathbf{m}_u^{p_u(t)})$ on the CPU
Transfer $\mathbf{w}_c^{p_u(t)}$ to the GPU
Predict $\hat{y}$ using Eq.**(30)**
return $\hat{y}$

Following OFA-FL, **OFA-style** is implemented as a once-for-all federated baseline that jointly trains a shared model across multiple nested subnetwork configurations. At deployment, the configuration matching the requested resource budget is selected directly. Unlike RAS-FL, these subnetworks follow a predefined width or capacity schedule rather than a learned federated structure-importance roadmap.

Slimmable-NN is implemented as a single shared network with switchable-width configurations, following the federated slimmable setting of FedSNN. During local training, clients sample from the supported widths, and the server aggregates the shared parameters. For evaluation, each shrinkage level is matched to the closest available width configuration. Unlike RAS-FL, Slimmable-NN adapts capacity through predefined width multipliers rather than learned, importance-guided structured masks.

SLEX-style, inspired by SLEXNet and ScaleFL , is implemented as an early-exit federated baseline by attaching intermediate classifiers to a shared backbone. All exits are trained jointly during local optimization. At deployment, the exit matching the target resource budget is selected, while the final exit represents the full-depth model. Because early exits reduce computation through depth-wise termination rather than channel-level shrinkage, each exit is compared using its measured active-parameter and FLOP profile.

We adapted all baselines to the same federated training and deployment setting used by RAS-FL. To the best of our knowledge, existing federated adaptations of these model families primarily address resource heterogeneity across clients or predefined deployment budgets, rather than reversible, training-free shrinking and expansion of the same deployed model as a client's available resources change. We therefore preserve each method's core adaptation mechanism while mapping it to the same runtime budgets and evaluation setting used by RAS-FL. All methods use identical client partitions, communication rounds, local epochs, optimizer settings, and evaluation clients.

1.3 Evaluation on Newly Introduced Clients

In addition to the standard global test set evaluation, we evaluate all methods on a separate set of *newly introduced clients* to better reflect realistic deployment scenarios.

In practice, FL systems continually receive new users whose data distributions were not observed during training. Therefore, evaluating only on the global test set does not fully capture the ability of a deployed model to generalize to unseen clients under non-IID data. To simulate this setting, we construct an additional group of clients using the original training dataset but with a different Dirichlet partition ($\alpha = 0.08$), producing substantially stronger label skew than the training clients. These clients are completely disjoint from the clients used during federated training and are never involved in model optimization. For each newly introduced client, we randomly split its local data into two disjoint subsets. The first subset is used as a small adaptation set, while the second is reserved exclusively for evaluation. The adaptation subset contains 500 samples and is used only by methods that support client-specific personalization. The evaluation subset also contains 500 samples and is never used during adaptation.

All baseline methods (FedAvg, FedProx, OFA-style, Slimmable-NN, SLEX-style, and FedDrop-style) are evaluated directly using the global model without additional client adaptation. RAS-FL supports two deployment modes:

- **Global roadmap**: all clients use the runtime roadmap learned during federated training.
- **Personalized roadmap**: each new client refines the global importance scores using its small adaptation subset to construct a client-specific runtime roadmap while keeping the shared model parameters unchanged.

This evaluation protocol measures the practical deployment performance of runtime-aware federated models on previously unseen clients while maintaining identical runtime budgets across all compared methods.

2 Additional Experimental Results

2.1 Extended Accuracy–Efficiency Results

Table 1 provides the complete results for Fashion-MNIST, CIFAR-10, CIFAR-100, and Tiny-ImageNet-20. In addition to full-model and average runtime performance, it reports the percentages of active parameters, structures, and FLOPs relative to the corresponding full model. RAS-FL maintains the strongest runtime accuracy on CIFAR-10, CIFAR-100, and Tiny-ImageNet-20, while remaining competitive on Fashion-MNIST. The results further show that its roadmap configurations provide substantial computational savings while preserving a favourable accuracy–efficiency trade-off. Global and personalised roadmaps exhibit similar resource usage because they follow the same structural budgets.

Figure 1 further shows that the client-level accuracy distributions of RAS-FL are shifted toward higher values on the more challenging datasets. Overall, the importance-guided roadmap reduces runtime cost while preserving predictive performance.

Accuracy–FLOP Trade-off Figure 2 complements the active-parameter analysis by comparing runtime accuracy with average active FLOPs. RAS-FL provides a favourable accuracy–computation trade-off across all datasets. The advantage is particularly clear on CIFAR-100 and Tiny-ImageNet-20, where RAS-FL substantially outperforms the adaptive baselines at comparable or lower computational cost. On Fashion-MNIST and CIFAR-10, it preserves high accuracy while reducing the active computation relative to the full model.

Table 1: Extended accuracy–efficiency results on previously unseen deployment clients. Full-model and average runtime performance are reported for each method, with runtime metrics averaged over the supported shrinkage levels from 5% to 40%. Active parameters, structures, and FLOPs are expressed as percentages of the corresponding full model. Values are mean $\pm$ standard deviation over three runs when available.

Dataset	Method	Full Acc. (%)	Avg. Runtime Acc. (%)	Macro F1 (%)	Active Params (%)	Active Struct. (%)	Active FLOPs (%)
F-MNIST	**FedAvg**	91.79 ± 0.74	91.47 ± 0.14	91.67 ± 0.32	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
	FedProx	91.83 ± 0.90	$\mathbf{91.78 \pm 0.67}$	$\mathbf{91.85 \pm 0.73}$	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
	Federated Dropout-style	87.76 ± 0.22	86.56 ± 0.42	87.72 ± 0.27	72.52 ± 0.00	81.19 ± 0.00	72.00 ± 0.00
	OFA-style	91.50 ± 0.49	91.05 ± 1.13	91.12 ± 1.18	82.72 ± 0.00	83.20 ± 0.00	82.72 ± 0.00
	SLEX-style final exit	91.20 ± 1.25	91.10 ± 1.66	91.25 ± 1.72	82.73 ± 0.00	83.86 ± 0.00	82.73 ± 0.00
	Slimmable Neural Networks	90.13 ± 1.68	89.90 ± 1.82	89.95 ± 1.64	68.85 ± 0.00	82.50 ± 0.00	69.13 ± 0.00
	RAS-FL global roadmap	$\mathbf{92.19 \pm 0.44}$	91.00 ± 1.87	91.79 ± 1.14	73.02 ± 0.14	82.86 ± 0.00	73.02 ± 0.14
	RAS-FL personalized roadmap	$\mathbf{92.19 \pm 0.44}$	91.02 ± 1.91	91.57 ± 1.44	72.88 ± 0.14	82.86 ± 0.00	73.02 ± 0.14
CIFAR-10	**FedAvg**	$\mathbf{83.64 \pm 2.08}$	$\mathbf{81.53 \pm 2.15}$	$\mathbf{83.58 \pm 2.16}$	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
	FedProx	81.17 ± 4.66	80.87 ± 4.76	81.03 ± 4.56	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
	Federated Dropout-style	30.30 ± 13.63	34.17 ± 13.35	31.05 ± 13.98	72.64 ± 0.00	82.63 ± 0.00	71.52 ± 0.00
	OFA-style	78.19 ± 3.57	76.01 ± 3.91	76.01 ± 3.91	71.66 ± 0.00	82.83 ± 0.00	69.12 ± 0.00
	SLEX-style final exit	74.22 ± 8.29	72.35 ± 5.78	74.15 ± 8.71	75.74 ± 0.00	83.18 ± 0.00	69.34 ± 0.00
	Slimmable Neural Networks	77.11 ± 1.43	78.14 ± 0.71	77.17 ± 0.41	68.82 ± 0.00	82.50 ± 0.00	69.00 ± 0.00
	RAS-FL global roadmap	81.44 ± 3.28	80.08 ± 2.79	81.38 ± 2.23	74.62 ± 0.00	82.63 ± 0.00	78.08 ± 0.00
	RAS-FL personalized roadmap	81.44 ± 3.28	79.85 ± 3.32	81.45 ± 3.38	74.62 ± 0.00	82.63 ± 0.00	78.08 ± 0.00
CIFAR-100	**FedAvg**	57.59 ± 5.71	56.37 ± 5.89	57.18 ± 5.76	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
	FedProx	49.73 ± 4.66	49.11 ± 4.62	49.79 ± 4.61	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
	Federated Dropout-style	11.84 ± 2.90	8.32 ± 0.72	8.32 ± 0.72	68.84 ± 0.36	82.50 ± 0.00	70.20 ± 0.55
	OFA-style	18.63 ± 7.67	14.15 ± 3.64	14.15 ± 3.64	67.26 ± 1.33	82.50 ± 0.00	67.28 ± 1.19
	SLEX-style final exit	24.02 ± 8.91	22.17 ± 14.62	22.17 ± 14.62	68.01 ± 2.07	82.50 ± 0.00	68.13 ± 1.05
	Slimmable Neural Networks	39.12 ± 6.25	40.75 ± 7.04	40.75 ± 7.04	68.79 ± 0.00	82.50 ± 0.00	68.88 ± 0.00
	RAS-FL global roadmap	$\mathbf{64.20 \pm 2.43}$	$\mathbf{54.95 \pm 1.08}$	$\mathbf{57.91 \pm 2.13}$	78.52 ± 0.28	82.50 ± 0.00	68.81 ± 0.47
	RAS-FL personalized roadmap	$\mathbf{64.20 \pm 2.43}$	49.22 ± 1.89	56.61 ± 1.40	78.52 ± 0.28	82.50 ± 0.00	68.81 ± 0.47
Tiny-IN20	**FedAvg**	33.10 ± 13.87	32.18 ± 14.53	33.81 ± 14.07	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
	FedProx	$\mathbf{43.47 \pm 1.31}$	$\mathbf{43.71 \pm 1.55}$	$\mathbf{43.45 \pm 1.26}$	100.00 ± 0.00	100.00 ± 0.00	100.00 ± 0.00
	Federated Dropout-style	5.64 ± 3.00	4.83 ± 2.85	4.83 ± 2.85	85.54 ± 0.00	82.50 ± 0.08	86.05 ± 0.03
	OFA-style	9.91 ± 3.39	12.27 ± 3.93	12.27 ± 3.93	86.83 ± 0.00	82.50 ± 0.00	87.30 ± 0.00
	SLEX-style final exit	13.46 ± 2.35	17.12 ± 6.40	17.12 ± 6.40	86.86 ± 0.00	82.50 ± 0.00	87.24 ± 0.13
	Slimmable Neural Networks	6.60 ± 1.98	7.49 ± 1.94	7.49 ± 1.94	86.24 ± 0.00	82.50 ± 0.00	86.53 ± 0.20
	RAS-FL global roadmap	38.33 ± 1.65	33.14 ± 3.13	33.14 ± 3.13	89.07 ± 0.00	82.50 ± 0.00	89.64 ± 0.07
	RAS-FL personalized roadmap	37.98 ± 2.04	32.97 ± 3.80	32.97 ± 3.80	89.07 ± 0.00	82.50 ± 0.00	89.64 ± 0.07

2.2 Training Convergence

Figure 3 shows that RAS-FL converges smoothly on Fashion-MNIST and CIFAR-10 and maintains a stable upward trend on CIFAR-100. Tiny-ImageNet exhibits larger round-to-round fluctuations due to its greater task complexity and non-IID client distributions; however, RAS-FL continues improving without divergence. Overall, shrinkage-aware training preserves reliable convergence while jointly preparing the full model and runtime submodels.

2.3 Shrinkage-Aware Training Dynamics

Figure 4 compares full-model training accuracy with masked-submodel validation accuracy as the supported mask schedule expands. Introducing additional shrinkage levels can temporarily reduce masked accuracy, particularly on the more challenging datasets, but performance subsequently recovers as the model adapts to the expanded runtime capacities. This indicates that shrinkage-aware training improves submodel robustness without destabilizing the full model.

2.4 Per-Level Runtime Adaptation Results

Table 2 shows that the useful shrinkage range is dataset- and architecture-dependent. Accuracy generally remains stable over moderate removal levels, after which a dataset-specific degradation point appears. Fashion-MNIST remains robust even under aggressive shrinkage, while CIFAR-10 preserves most of its accuracy through approximately 30% removal. The more challenging CIFAR-100 and Tiny-ImageNet-20 set-

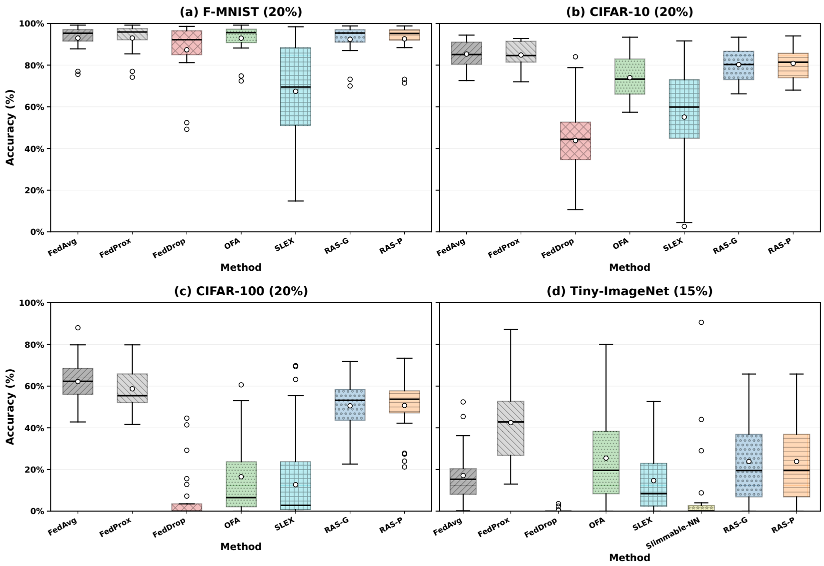

Figure 1: Client-level runtime accuracy at representative shrinkage levels: 20% for Fashion-MNIST and the CIFAR datasets, and 15% for Tiny-ImageNet-200.

tings exhibit earlier and sharper degradation, reflecting their greater task and representation complexity. For CIFAR-100, the global roadmap retains $51.60 \pm 2.38\%$ accuracy at 30% structure removal, while physical compaction reduces parameters by $58.61 \pm 0.17\%$, FLOPs by $53.49 \pm 2.01\%$, compact-state size by $58.60 \pm 0.17\%$, and model memory by $56.49 \pm 0.34\%$. The larger drop at 40% suggests that this shrinkage level removes more capacity than the trained full model can tolerate on CIFAR-100.Therefore, the roadmap should be viewed as a set of operating points from which the runtime policy selects an appropriate accuracy–resource trade-off, rather than requiring every shrinkage level to preserve the same accuracy. Importantly, switching between operating points requires neither retraining nor fine-tuning. The selected compact model is reconstructed from the fixed full model and its precomputed roadmap, keeping runtime adaptation lightweight and reversible.

Sensitivity to checkpoint-selection weights. Table 5 shows that Fashion-MNIST is relatively insensitive to the checkpoint weights, although ignoring masked-submodel performance, i.e., $(1.00, 0.00)$, yields the lowest average runtime accuracy. CIFAR-10 is more sensitive, with the balanced setting $(0.50, 0.50)$ providing the highest and most stable accuracy across all shrinkage levels. In contrast, assigning greater weight to the masked-submodel term, $(0.25, 0.75)$, substantially reduces accuracy and increases variance. This suggests that, for the more challenging CIFAR-10 task, the full-model term should not be underweighted. Because the full model provides the shared representation from which all compact subnetworks are derived, overemphasising masked performance may favour checkpoints specialised to sampled subnetworks at the expense of general representation quality. Overall, the results support balanced consideration of full-model and masked-submodel accuracy during checkpoint selection.

Sensitivity to the number of training shrinkage levels. Table 6 evaluates the number of shrinkage levels sampled during shrinkage-aware training. Overall, RAS-FL

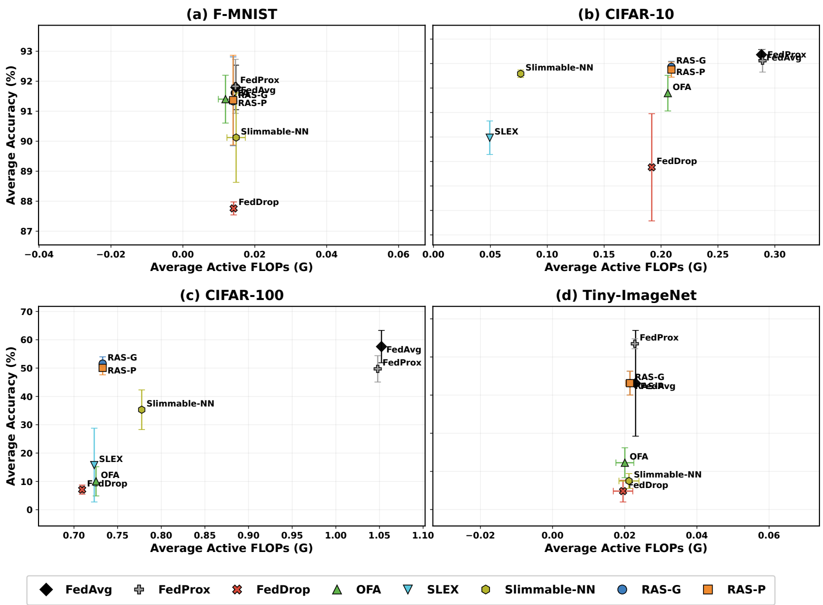

Figure 2: Runtime accuracy versus average active FLOPs on previously unseen clients. Error bars indicate standard deviation over three independent runs.

shows limited sensitivity to this hyperparameter. On Fashion-MNIST, the full-model accuracy varies by only 0.21 percentage points across all configurations. On CIFAR-10, the variation is slightly larger, with the best and worst configurations differing by about one percentage point. Although the two- and five-level settings perform marginally better on Fashion-MNIST and the three-level setting achieves the highest accuracy on CIFAR-10, the seven-level configuration used by RAS-FL remains competitive while providing finer coverage of runtime capacities. These results indicate that increasing the number of training shrinkage levels incurs only a small reduction in full-model accuracy.

Sensitivity to the mask-training probability. Table 7 shows that the effect of p_{mask} is task dependent. On Fashion-MNIST, lower masking probabilities perform best: $p_{\mathrm{mask}} = 0.10$ achieves the highest accuracy, macro-F1, balanced accuracy, and checkpoint score, while $p_{\mathrm{mask}} = 0.25$ provides nearly identical performance. Increasing p_{mask} to 0.50 or 0.75 progressively reduces accuracy and leads to earlier checkpoint selection, suggesting that excessive masked updates can weaken full-model optimisation.

CIFAR-10 benefits from a moderate masking frequency. $p_{\mathrm{mask}} = 0.25$ provides the highest test accuracy and lowest loss, whereas $p_{\mathrm{mask}} = 0.50$ achieves the highest checkpoint score with low variance. By contrast, $p_{\mathrm{mask}} = 0.10$ exhibits greater variability, suggesting insufficient exposure to compact submodels, while $p_{\mathrm{mask}} = 0.75$ reduces predictive performance. Overall, values between 0.25 and 0.50 provide the best balance between full-model learning and robustness across compact subnetworks.

Sensitivity to the importance-score EMA coefficient. Table 8 shows that Fashion-MNIST is relatively insensitive to β, with all settings achieving similar predictive per-

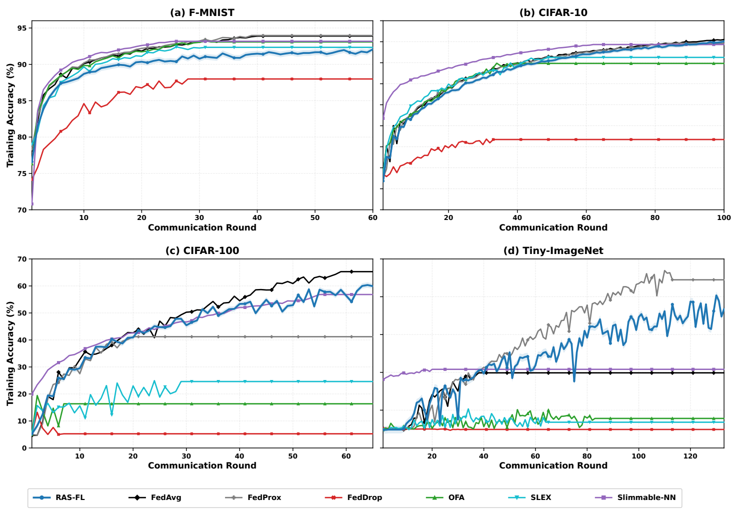

Figure 3: Training convergence of RAS-FL and competing baselines over communication rounds. Despite jointly training the full model and runtime-adaptive submodels, RAS-FL exhibits stable convergence across datasets.

formance. The setting $\beta = 0.95$ provides the highest accuracy, macro-F1, and balanced accuracy, whereas $\beta = 0.90$ gives the highest checkpoint score and $\beta = 0.99$ the lowest test loss. On CIFAR-10, larger EMA coefficients generally provide more stable importance estimates: $\beta = 0.99$ achieves the highest accuracy with low variance, while $\beta = 0.95$ obtains the highest checkpoint score. In contrast, $\beta = 0.90$ substantially reduces accuracy and increases variability. Overall, $\beta \in \{0.95, 0.99\}$ provides the most reliable balance between smoothing short-term importance fluctuations and remaining responsive to changes during federated training.

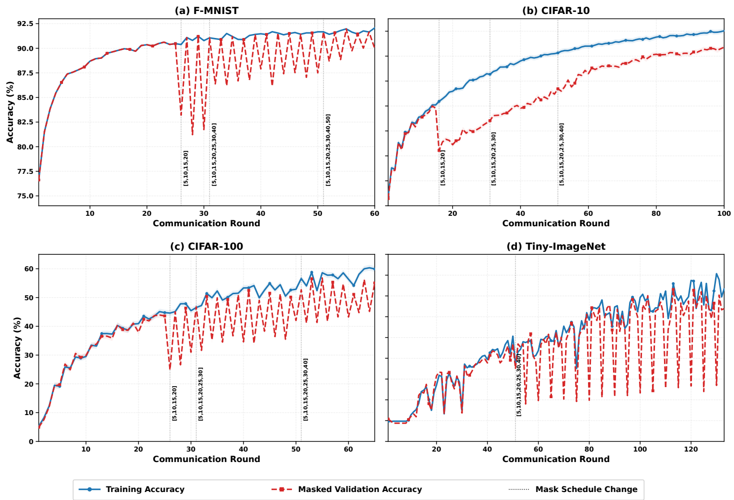

Figure 4: Shrinkage-aware training dynamics across communication rounds. Vertical lines mark expansions of the mask schedule, while the curves compare full-model training accuracy with masked-submodel validation accuracy.

Table 2: Runtime accuracy (%) at different shrinkage levels on previously unseen clients. Values are mean $\pm$ standard deviation over three runs. Fashion-MNIST, CIFAR-10, and CIFAR-100 are evaluated up to 50% structure removal, whereas Tiny-ImageNet-20 is evaluated up to 30%. Non-adaptive full-model baselines are reported only at 0%.

Dataset	Method	0%	10%	20%	30%	40%	50%
F-MNIST	**FedAvg**	91.79 ± 0.74	N/A	N/A	N/A	N/A	N/A
	FedProx	91.83 ± 0.90	N/A	N/A	N/A	N/A	N/A
	Federated Dropout-style	87.76 ± 0.22	87.02 ± 3.47	86.81 ± 1.15	84.33 ± 1.60	83.56 ± 2.82	83.92 ± 1.24
	OFA-style	91.50 ± 0.49	$\mathbf{91.58 \pm 0.62}$	90.82 ± 1.48	90.43 ± 1.51	89.82 ± 1.03	88.51 ± 1.40
	SLEX-style final exit	91.20 ± 1.25	91.27 ± 1.62	$\mathbf{91.05 \pm 1.79}$	$\mathbf{90.88 \pm 1.74}$	89.49 ± 2.80	87.02 ± 4.22
	Slimmable-NN	90.13 ± 1.68	90.17 ± 1.43	90.05 ± 1.80	89.22 ± 2.80	89.37 ± 1.75	88.74 ± 2.12
	RAS-FL global roadmap	$\mathbf{92.19 \pm 0.44}$	91.00 ± 2.06	90.93 ± 1.98	90.71 ± 2.03	90.26 ± 1.43	$\mathbf{89.97 \pm 0.25}$
	RAS-FL personalized roadmap	$\mathbf{92.19 \pm 0.44}$	91.03 ± 2.12	91.03 ± 2.00	90.61 ± 2.14	$\mathbf{90.36 \pm 1.17}$	89.26 ± 0.33
CIFAR-10	**FedAvg**	$\mathbf{83.64 \pm 2.08}$	N/A	N/A	N/A	N/A	N/A
	FedProx	81.17 ± 4.66	N/A	N/A	N/A	N/A	N/A
	Federated Dropout-style	30.30 ± 13.63	34.07 ± 7.80	36.97 ± 16.37	37.63 ± 21.90	31.12 ± 16.54	17.12 ± 5.65
	OFA-style	78.19 ± 3.57	80.14 ± 2.73	77.17 ± 4.01	67.92 ± 7.28	54.59 ± 4.58	27.59 ± 5.36
	SLEX-style final exit	74.22 ± 8.29	75.60 ± 6.77	73.37 ± 5.16	65.94 ± 4.89	47.22 ± 5.66	24.79 ± 9.99
	Slimmable-NN	77.11 ± 1.43	78.40 ± 1.23	78.97 ± 1.07	75.85 ± 1.51	65.08 ± 1.32	57.93 ± 4.01
	RAS-FL global roadmap	81.44 ± 3.28	80.74 ± 3.44	$\mathbf{80.38 \pm 2.87}$	$\mathbf{78.68 \pm 2.25}$	$\mathbf{72.12 \pm 2.10}$	$\mathbf{69.02 \pm 1.96}$
	RAS-FL personalized roadmap	81.44 ± 3.28	$\mathbf{80.86 \pm 3.59}$	79.92 ± 3.53	77.55 ± 3.11	71.62 ± 3.03	67.66 ± 3.36
CIFAR-100	**FedAvg**	57.59 ± 5.71	N/A	N/A	N/A	N/A	N/A
	FedProx	49.73 ± 4.66	N/A	N/A	N/A	N/A	N/A
	Federated Dropout-style	11.84 ± 2.90	9.64 ± 0.60	8.85 ± 0.37	7.09 ± 1.61	6.68 ± 1.74	5.70 ± 3.00
	OFA-style	18.63 ± 7.67	16.75 ± 5.20	13.03 ± 2.67	10.03 ± 5.15	5.14 ± 7.86	2.41 ± 2.09
	SLEX-style final exit	24.02 ± 8.91	26.70 ± 14.28	23.88 ± 20.90	12.27 ± 9.01	8.38 ± 5.33	7.72 ± 3.88
	Slimmable-NN	39.12 ± 6.25	39.89 ± 8.16	43.76 ± 8.33	35.30 ± 7.01	$\mathbf{21.11 \pm 10.12}$	$\mathbf{10.77 \pm 6.65}$
	RAS-FL global roadmap	$\mathbf{64.20 \pm 2.43}$	$\mathbf{60.88 \pm 2.13}$	$\mathbf{60.14 \pm 1.48}$	$\mathbf{51.60 \pm 2.38}$	17.78 ± 8.43	7.81 ± 1.53
	RAS-FL personalized roadmap	$\mathbf{64.20 \pm 2.43}$	60.70 ± 2.16	59.66 ± 2.31	50.07 ± 2.44	17.61 ± 5.68	5.15 ± 1.70
Tiny-IN20	**FedAvg**	33.10 ± 13.87	N/A	N/A	N/A	N/A	N/A
	FedProx	$\mathbf{43.47 \pm 1.31}$	N/A	N/A	N/A	N/A	N/A
	Federated Dropout-style	5.64 ± 3.00	4.43 ± 3.41	6.22 ± 3.51	3.01 ± 2.36	N/A	N/A
	OFA-style	16.54 ± 7.85	12.34 ± 2.57	10.28 ± 2.73	9.91 ± 3.39	N/A	N/A
	SLEX-style final exit	20.08 ± 12.02	18.19 ± 7.02	13.46 ± 2.35	12.74 ± 5.48	N/A	N/A
	Slimmable-NN	9.09 ± 0.62	7.28 ± 3.14	6.97 ± 2.56	6.60 ± 1.98	N/A	N/A
	RAS-FL global roadmap	38.33 ± 1.65	$\mathbf{37.00 \pm 3.50}$	$\mathbf{34.52 \pm 3.09}$	$\mathbf{22.70 \pm 7.78}$	N/A	N/A
	RAS-FL personalized roadmap	37.98 ± 2.04	36.00 ± 4.02	32.52 ± 5.56	21.60 ± 6.43	N/A	N/A

Table 3: Physical compression and server-GPU runtime of compact RAS-FL across 20 clients. Values are mean ± standard deviation over three independent runs.

Data	Variant	Metric	Full	10%	20%	30%	40%
F-MNIST	Global	**Accuracy (%)**	91.62 ± 0.01	90.88 ± 0.67	90.91 ± 0.73	90.80 ± 0.76	89.97 ± 0.25
		Parameter reduction (%)	0.00 ± 0.00	14.71 ± 0.00	32.44 ± 0.00	50.21 ± 0.05	67.63 ± 0.00
		FLOPs reduction (%)	0.00 ± 0.00	14.62 ± 0.00	17.95 ± 0.00	24.90 ± 0.22	44.45 ± 0.00
		Compact-state reduction (%)	0.00 ± 0.00	14.71 ± 0.00	32.43 ± 0.00	50.20 ± 0.05	67.62 ± 0.00
		Model-memory reduction (%)	0.00 ± 0.00	6.10 ± 0.00	32.39 ± 0.00	50.14 ± 0.04	67.54 ± 0.00
		Peak-memory reduction (%)	0.00 ± 0.00	1.30 ± 0.00	6.91 ± 0.00	10.70 ± 0.01	14.44 ± 0.00
	Personalised	**Accuracy (%)**	91.62 ± 0.01	90.92 ± 0.73	91.14 ± 0.61	91.24 ± 0.73	89.91 ± 0.52
		Parameter reduction (%)	0.00 ± 0.00	14.71 ± 0.00	32.43 ± 0.01	50.07 ± 0.03	67.62 ± 0.21
		FLOPs reduction (%)	0.00 ± 0.00	14.62 ± 0.00	17.97 ± 0.04	24.69 ± 0.43	44.46 ± 0.29
		Compact-state reduction (%)	0.00 ± 0.00	14.71 ± 0.00	32.42 ± 0.01	50.06 ± 0.03	67.61 ± 0.21
		Model-memory reduction (%)	0.00 ± 0.00	6.10 ± 0.00	32.38 ± 0.01	50.00 ± 0.04	67.53 ± 0.21
		Peak-memory reduction (%)	0.00 ± 0.00	1.30 ± 0.00	6.91 ± 0.00	10.67 ± 0.01	14.44 ± 0.04
CIFAR-10	Global	**Accuracy (%)**	82.57 ± 1.85	82.70 ± 2.02	81.56 ± 0.86	81.21 ± 0.23	75.06 ± 2.81
		Parameter reduction (%)	0.00 ± 0.00	19.16 ± 1.21	40.23 ± 1.43	57.46 ± 2.17	68.71 ± 0.08
		FLOPs reduction (%)	0.00 ± 0.00	15.22 ± 0.81	26.54 ± 2.02	34.11 ± 1.41	42.81 ± 0.17
		Compact-state reduction (%)	0.00 ± 0.00	19.15 ± 1.21	40.22 ± 1.43	57.45 ± 2.17	68.69 ± 0.08
		Model-memory reduction (%)	0.00 ± 0.00	20.06 ± 0.11	41.65 ± 1.37	54.69 ± 1.45	69.53 ± 1.19
		Peak-memory reduction (%)	0.00 ± 0.00	5.94 ± 0.02	11.29 ± 0.44	15.17 ± 0.36	18.73 ± 0.82
	Personalised	**Accuracy (%)**	82.57 ± 1.85	82.46 ± 1.68	82.02 ± 1.25	80.15 ± 0.26	74.79 ± 0.50
		Parameter reduction (%)	0.00 ± 0.00	18.86 ± 1.02	39.62 ± 1.28	57.34 ± 1.44	68.69 ± 0.07
		FLOPs reduction (%)	0.00 ± 0.00	14.40 ± 1.25	25.98 ± 1.27	34.90 ± 0.93	43.23 ± 0.01
		Compact-state reduction (%)	0.00 ± 0.00	18.86 ± 1.02	39.61 ± 1.28	57.33 ± 1.44	68.67 ± 0.07
		Model-memory reduction (%)	0.00 ± 0.00	19.02 ± 0.75	41.27 ± 1.23	54.80 ± 1.79	69.51 ± 1.20
		Peak-memory reduction (%)	0.00 ± 0.00	5.52 ± 0.07	10.53 ± 0.49	15.35 ± 0.65	18.73 ± 0.82
CIFAR-100	Global	**Accuracy (%)**	64.20 ± 2.43	60.88 ± 2.13	60.14 ± 1.48	51.60 ± 2.38	17.78 ± 8.43
		Parameter reduction (%)	0.00 ± 0.00	19.00 ± 0.44	40.32 ± 0.90	58.61 ± 0.17	66.88 ± 0.03
		FLOPs reduction (%)	0.00 ± 0.00	22.42 ± 0.59	38.67 ± 0.07	53.49 ± 2.01	63.70 ± 0.31
		Compact-state reduction (%)	0.00 ± 0.00	19.00 ± 0.44	40.31 ± 0.90	58.60 ± 0.17	66.87 ± 0.03
		Model-memory reduction (%)	0.00 ± 0.00	22.79 ± 0.42	43.07 ± 0.86	56.49 ± 0.34	68.39 ± 0.03
		Peak-memory reduction (%)	0.00 ± 0.00	3.97 ± 0.00	13.85 ± 0.88	20.76 ± 0.99	24.44 ± 0.73
	Personalised	**Accuracy (%)**	64.20 ± 2.43	60.70 ± 2.16	59.66 ± 2.31	50.07 ± 2.44	17.61 ± 5.68
		Parameter reduction (%)	0.00 ± 0.00	18.86 ± 0.43	40.18 ± 0.60	58.69 ± 0.11	66.88 ± 0.02
		FLOPs reduction (%)	0.00 ± 0.00	22.32 ± 0.62	38.54 ± 0.23	52.58 ± 1.28	63.70 ± 0.20
		Compact-state reduction (%)	0.00 ± 0.00	18.86 ± 0.43	40.17 ± 0.60	58.68 ± 0.11	66.87 ± 0.02
		Model-memory reduction (%)	0.00 ± 0.00	22.66 ± 0.41	42.94 ± 0.57	56.35 ± 0.23	68.39 ± 0.02
		Peak-memory reduction (%)	0.00 ± 0.00	4.60 ± 0.39	14.17 ± 1.02	20.58 ± 0.86	24.22 ± 1.01

Table 4: Complete ablation accuracy (%) on previously unseen Fashion-MNIST and CIFAR-10 clients. Values are mean ± standard deviation over three runs. The highest value in each column is shown in bold with grey shading.

Dataset	Variant	Full	5%	10%	15%	20%	25%	30%	40%
F-MNIST	**Full RAS-FL**	92.03 ± 0.23	91.43 ± 1.32	90.92 ± 2.21	90.89 ± 2.08	90.90 ± 2.03	**91.18 ± 1.99**	**91.04 ± 1.74**	**90.52 ± 0.69**
	Random masks	91.91 ± 1.01	91.47 ± 1.35	90.93 ± 1.67	90.03 ± 1.22	87.61 ± 4.52	88.18 ± 2.60	87.38 ± 1.75	79.96 ± 3.28
	No distillation	92.01 ± 0.91	91.25 ± 1.19	**91.23 ± 1.17**	**91.21 ± 1.22**	**91.19 ± 1.19**	90.10 ± 1.31	89.84 ± 1.42	87.52 ± 2.02
	No shrink training	92.04 ± 0.72	91.02 ± 0.80	89.52 ± 2.40	89.11 ± 2.50	88.16 ± 2.98	86.67 ± 3.05	84.52 ± 3.40	76.52 ± 5.83
	No runtime selection	**92.31 ± 0.80**	**91.78 ± 1.48**	91.15 ± 2.32	91.08 ± 2.30	90.98 ± 2.36	91.09 ± 2.46	90.91 ± 2.13	89.87 ± 2.20
CIFAR-10	**Full RAS-FL**	82.67 ± 2.02	**82.60 ± 2.03**	**82.36 ± 2.30**	**81.83 ± 2.07**	**81.16 ± 2.15**	**81.02 ± 2.07**	**80.21 ± 3.38**	**72.71 ± 4.01**
	Random masks	72.63 ± 11.75	70.09 ± 14.09	63.77 ± 23.13	59.54 ± 22.11	50.01 ± 20.37	41.23 ± 15.09	38.16 ± 11.49	20.13 ± 6.46
	No distillation	**82.95 ± 2.10**	82.57 ± 2.33	82.33 ± 2.16	81.71 ± 2.03	80.34 ± 2.20	80.80 ± 2.23	74.72 ± 3.10	66.98 ± 2.51
	No shrink training	82.31 ± 2.89	81.05 ± 2.30	79.88 ± 3.16	77.79 ± 4.16	75.48 ± 4.18	72.99 ± 4.70	68.65 ± 4.36	58.27 ± 4.67
	No runtime selection	80.09 ± 4.69	79.87 ± 5.15	79.52 ± 4.91	79.00 ± 4.56	78.01 ± 4.98	76.56 ± 4.76	75.11 ± 4.39	63.92 ± 5.22

Table 5: Sensitivity analysis of the runtime-aware checkpoint weights on newly introduced Fashion-MNIST and CIFAR-10 clients. Values are mean ± standard deviation over three independent runs.

Data	$(\lambda_{\mathrm{full}}, \lambda_{\mathrm{mask}})$	Full Acc.	Runtime Avg.	10%	20%	30%	40%	Active Params	Active Struct.	Active FLOPs	Runs
F-MNIST	$(\mathbf{1.00}, \mathbf{0.00})$	92.90 ± 0.37	92.31 ± 0.29	92.53 ± 0.20	92.54 ± 0.20	92.29 ± 0.38	91.90 ± 0.60	64.39 ± 0.14	75.51 ± 0.00	64.39 ± 0.14	3
	$(\mathbf{0.75}, \mathbf{0.25})$	**92.93 ± 0.38**	**92.38 ± 0.27**	**92.61 ± 0.27**	92.49 ± 0.29	92.34 ± 0.35	**92.08 ± 0.65**	64.39 ± 0.14	75.51 ± 0.00	64.39 ± 0.14	3
	$(\mathbf{0.50}, \mathbf{0.50})$	**92.93 ± 0.38**	92.37 ± 0.26	**92.61 ± 0.25**	92.54 ± 0.33	**92.40 ± 0.35**	91.95 ± 0.46	64.39 ± 0.13	75.51 ± 0.00	64.39 ± 0.13	3
	$(\mathbf{0.25}, \mathbf{0.75})$	**92.93 ± 0.38**	92.36 ± 0.30	**92.61 ± 0.27**	**92.56 ± 0.31**	92.34 ± 0.39	91.93 ± 0.72	64.40 ± 0.14	75.51 ± 0.00	64.40 ± 0.14	3
CIFAR-10	$(\mathbf{1.00}, \mathbf{0.00})$	81.05 ± 1.09	77.31 ± 1.26	80.55 ± 1.22	79.89 ± 0.82	77.59 ± 1.05	71.22 ± 2.43	70.91 ± 0.93	75.18 ± 0.00	70.91 ± 0.93	3
	$(\mathbf{0.75}, \mathbf{0.25})$	77.77 ± 4.30	74.73 ± 3.93	77.06 ± 3.97	77.52 ± 3.34	74.61 ± 4.17	69.74 ± 4.65	73.17 ± 2.66	75.18 ± 0.00	73.17 ± 2.66	3
	$(\mathbf{0.50}, \mathbf{0.50})$	**82.16 ± 0.26**	**79.29 ± 1.27**	**81.54 ± 0.67**	**81.31 ± 0.71**	**79.85 ± 1.07**	**74.44 ± 2.82**	69.93 ± 0.22	75.18 ± 0.00	69.93 ± 0.22	3
	$(\mathbf{0.25}, \mathbf{0.75})$	74.38 ± 12.51	67.25 ± 21.32	74.98 ± 11.25	71.33 ± 16.91	65.18 ± 25.68	57.53 ± 31.42	73.40 ± 5.73	75.18 ± 0.00	73.40 ± 5.73	3

Table 6: Sensitivity to the number of shrinkage levels sampled during shrinkage-aware training on Fashion-MNIST and CIFAR-10. Values are mean ± standard deviation over three independent runs. The best full-model accuracy within each dataset is shown in bold.

Data	No.	Training shrinkage levels (%)	Full Acc. (%)	Selected Round
F-MNIST	2	$\{10, 40\}$	$\mathbf{91.96 \pm 0.19}$	49.67 ± 3.06
	3	$\{10, 25, 40\}$	91.75 ± 0.16	43.67 ± 5.03
	5	$\{5, 10, 20, 30, 40\}$	91.95 ± 0.04	49.67 ± 7.02
	7	$\{5, 10, 15, 20, 25, 30, 40\}$	91.80 ± 0.05	45.00 ± 2.00
CIFAR-10	2	$\{10, 40\}$	79.35 ± 1.25	92.00 ± 10.39
	3	$\{10, 25, 40\}$	$\mathbf{80.45 \pm 0.28}$	99.33 ± 1.15
	5	$\{5, 10, 20, 30, 40\}$	80.30 ± 0.18	99.33 ± 1.15
	7	$\{5, 10, 15, 20, 25, 30, 40\}$	79.45 ± 1.41	90.00 ± 13.00

Table 7: Sensitivity analysis of the mask-training probability p_{mask} on Fashion-MNIST and CIFAR-10 using their corresponding CNN architectures. Values are mean ± standard deviation over three independent runs. Accuracy, macro-F1, balanced accuracy, and checkpoint score are reported as percentages.

Data	p_{mask}	Test Acc.	Macro-F1	Balanced Acc.	Test Loss	Checkpoint Score	Best Round	Runs
F-MNIST	0.10	$\mathbf{91.95 \pm 0.08}$	$\mathbf{91.93 \pm 0.05}$	$\mathbf{91.82 \pm 0.12}$	$\mathbf{0.2294 \pm 0.0017}$	$\mathbf{92.58 \pm 0.27}$	49.67 ± 3.06	3
	0.25	91.93 ± 0.17	91.92 ± 0.54	91.63 ± 0.33	0.2299 ± 0.0030	92.52 ± 0.30	48.33 ± 3.06	3
	0.50	91.38 ± 0.59	91.36 ± 0.28	90.78 ± 0.09	0.2386 ± 0.0135	92.21 ± 0.31	39.67 ± 14.47	3
	0.75	90.99 ± 0.33	90.96 ± 0.31	90.23 ± 0.29	0.2500 ± 0.0037	91.92 ± 0.13	24.00 ± 1.00	3
CIFAR-10	0.10	79.46 ± 3.05	79.44 ± 3.03	78.60 ± 2.82	0.5981 ± 0.0847	73.85 ± 5.03	86.00 ± 22.54	3
	0.25	$\mathbf{79.55 \pm 0.73}$	$\mathbf{79.54 \pm 0.71}$	$\mathbf{79.02 \pm 0.69}$	$\mathbf{0.5929 \pm 0.0201}$	76.43 ± 0.73	91.67 ± 8.50	3
	0.50	79.21 ± 0.35	79.18 ± 0.31	78.29 ± 0.21	0.6058 ± 0.0040	$\mathbf{77.31 \pm 0.43}$	98.67 ± 1.15	3
	0.75	77.49 ± 0.57	77.45 ± 0.55	77.40 ± 0.82	0.6563 ± 0.0186	76.65 ± 0.24	99.00 ± 1.00	3

Table 8: Sensitivity analysis of the importance-score exponential moving-average coefficient β on Fashion-MNIST and CIFAR-10. Values are mean ± standard deviation over three independent runs. Accuracy, macro-F1, balanced accuracy, and checkpoint score are reported as percentages.

Data	EMA β	Test Acc.	Macro-F1	Balanced Acc.	Test Loss	Checkpoint Score	Best Round	Runs
F-MNIST	0.80	91.83 ± 0.03	91.15 ± 0.21	90.73 ± 0.27	0.2298 ± 0.0026	92.59 ± 0.12	49.67 ± 8.33	3
	0.90	91.83 ± 0.05	91.15 ± 0.22	90.77 ± 0.19	0.2297 ± 0.0022	$\mathbf{92.61 \pm 0.12}$	49.00 ± 7.21	3
	0.95	$\mathbf{91.93 \pm 0.17}$	$\mathbf{91.89 \pm 0.25}$	$\mathbf{91.23 \pm 0.11}$	0.2299 ± 0.0030	92.52 ± 0.30	48.33 ± 3.06	3
	0.99	91.89 ± 0.13	91.91 ± 0.06	90.91 ± 0.41	$\mathbf{0.2292 \pm 0.0042}$	92.52 ± 0.15	46.33 ± 3.06	3
CIFAR-10	0.80	79.47 ± 1.20	79.49 ± 1.26	76.21 ± 1.29	$\mathbf{0.5770 \pm 0.0166}^{\dagger}$	76.24 ± 1.24	91.33 ± 9.02	3
	0.90	75.57 ± 5.93	75.44 ± 5.66	74.87 ± 4.36	$0.7372 \pm 0.2080^{\dagger}$	71.38 ± 7.93	69.67 ± 29.50	3
	0.95	79.55 ± 0.73	79.57 ± 0.62	77.78 ± 0.98	$0.5820 \pm 0.0097^{\dagger}$	$\mathbf{76.43 \pm 0.73}$	91.67 ± 8.50	3
	0.99	$\mathbf{79.93 \pm 0.45}$	$\mathbf{79.96 \pm 0.42}$	$\mathbf{79.03 \pm 0.25}$	$0.5906 \pm 0.0187^{\dagger}$	76.06 ± 0.50	96.00 ± 5.29	3